

# STREETWISE: SMARTER STORMWATER MANAGEMENT

City of Savannah | Innovation Team

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## Keywords:

API Integration, Service Request Management, Optimization

## Summary:

StreetWise is a web-based platform built for the City of Savannah's stormwater team to centralize case history, crew activity, and operational data that currently lives across disconnected systems. Using ArcGIS Experience Builder, custom **React** widgets, and **automated** data pipelines, the tool integrates real-time sensor readings, 311 service requests, and city GIS layers into a single interactive interface.

# STREETWISE: SMARTER STORMWATER MANAGEMENT

Innovation Team

Manuel Dominguez — Team Director

coding it forward >



ANDREW  
ADISOEMARTA

# ABOUT ME

- MSCS at UT Austin
- Software Fellow with the City of Savannah's Innovation Team



Savannah currently experiences approx. **10–15 flood-triggering storm events** per year.  
Frequency could increase by **15–20% by 2037.**

[NOAA Atlas 14 1-year/1-hour threshold for coastal Georgia](#)

[Chiappa et al., 2024, \*Geophysical Research Letters\*; 5th National Climate Assessment, Southeast chapter](#)

Savannah's Stormwater team does critical work every day — but the knowledge of what was done, where, and why has been scattered across spreadsheets, emails, and memory

# BACKGROUND

- Stormwater team workflow is reactive
- Work orders, 311 cases, field notes, crew activity, and project history all live in different places
- Institutional knowledge is lost with staff turnover

# VISION

- One centralized platform where all stormwater activity is logged, searchable, and visible
- No more scrambling between 4+ different applications
- Connects the office and the field in real time
- Builds a lasting record the whole team can access and build on

# RESEARCH

- Comprehensive journey maps and storyboards were developed to assist in identifying what features to integrate

7

PROCESS PHASES

4+

DISCONNECTED SYSTEMS

10+

STAKEHOLDER TYPES

5

SPECIALIZED CREWS

~30

DAYS AVG. SLA

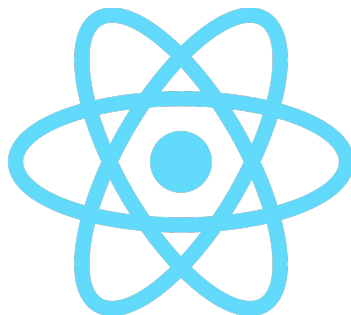
12+

PAIN POINTS

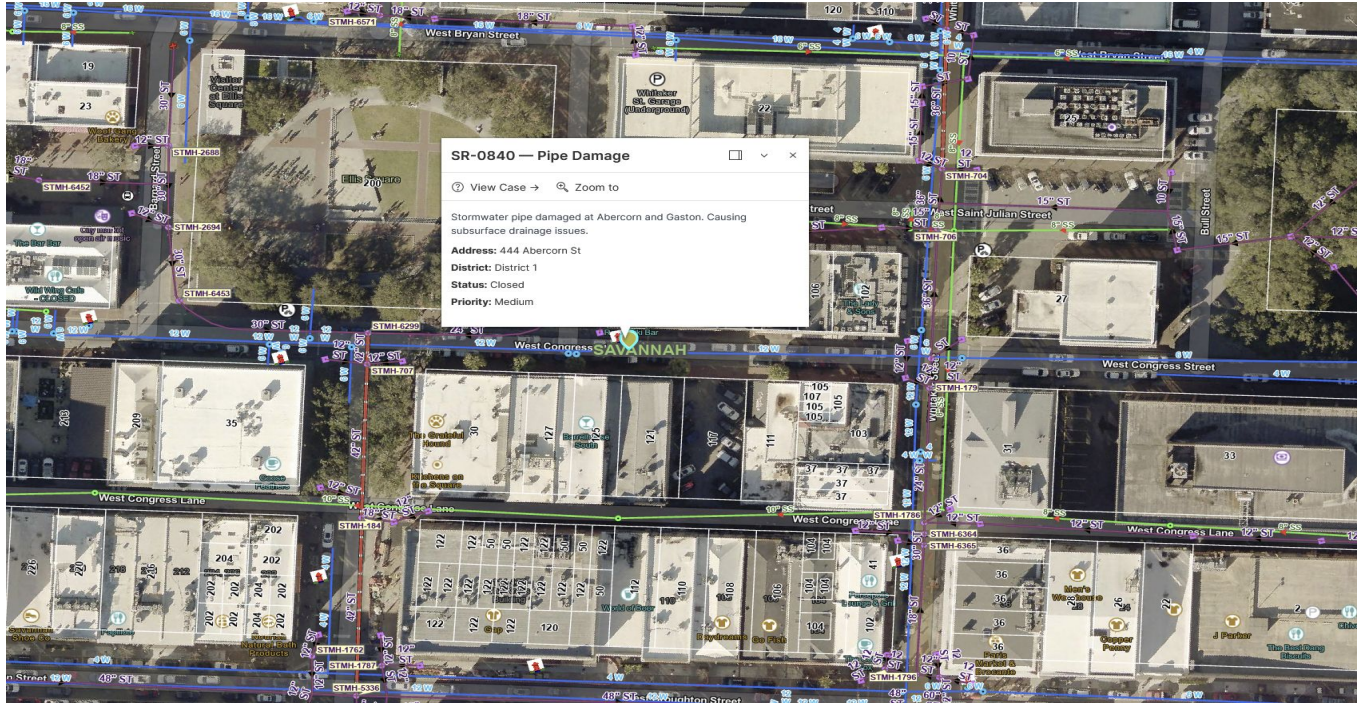


# BUILDING FOR LONGEVITY

- Started with UI screens and developed in to prototype
- Built in ArcGIS Experience Builder instead of standalone app
- Not a one and done prototype, but lives past my Fellowship
- Custom widgets built in React



# KEY FEATURES - LIVE MAP



# KEY FEATURES - SERVICE REQUEST MANAGEMENT

## Service Requests

8 total - 3 high priority

[+ New Request](#)

Open ▾ In Progress ▾ Closed ▾ High ▾ Medium ▾ Low ▾

| ID                                               | TYPE                                        | LOCATION                        | STATUS      | PRIORITY | ASSIGNED                    | CREATED         | SLA     |
|--------------------------------------------------|---------------------------------------------|---------------------------------|-------------|----------|-----------------------------|-----------------|---------|
| <input type="checkbox"/> <a href="#">SR-0847</a> | Clogged Storm Drain<br><a href="#">311</a>  | 123 E Bay St<br>District 1      | In Progress | ● High   | <a href="#">JS</a> J. Smith | Jun 2, 2:30 PM  | 2h left |
| <input type="checkbox"/> <a href="#">SR-0846</a> | Street Flooding<br><a href="#">311</a>      | 456 W Harris St<br>District 3   | Open        | ● High   | Unassigned                  | Jun 2, 1:15 PM  | 4h left |
| <input type="checkbox"/> <a href="#">SR-0845</a> | Ditch Debris<br><a href="#">SMARTSHEET</a>  | 789 Montgomery St<br>District 2 | In Progress | ● Medium | <a href="#">MJ</a> M. Jones | Jun 1, 4:00 PM  | 1d left |
| <input type="checkbox"/> <a href="#">SR-0844</a> | Manhole Issue<br><a href="#">INTERNAL</a>   | 321 Bull St<br>District 1       | Closed      | ● Low    | <a href="#">RD</a> R. Davis | May 30, 9:00 AM | Met     |
| <input type="checkbox"/> <a href="#">SR-0843</a> | Canal Backup<br><a href="#">311</a>         | 555 Waters Ave<br>District 5    | Open        | ● High   | Unassigned                  | Jun 2, 10:45 AM | 6h left |
| <input type="checkbox"/> <a href="#">SR-0842</a> | Inlet Blocked<br><a href="#">SMARTSHEET</a> | 888 Drayton St<br>District 4    | In Progress | ● Medium | <a href="#">JS</a> J. Smith | Jun 1, 11:30 AM | 8h left |
| <input type="checkbox"/> <a href="#">SR-0841</a> | Ditch Mowing<br><a href="#">INTERNAL</a>    | 222 Liberty St<br>District 2    | Open        | ● Low    | Unassigned                  | May 31, 3:20 PM | 3d left |
| <input type="checkbox"/> <a href="#">SR-0840</a> | Pipe Damage<br><a href="#">SMARTSHEET</a>   | 444 Abercorn St<br>District 1   | Closed      | ● Medium | <a href="#">MJ</a> M. Jones | May 28, 8:00 AM | Met     |

# NEXT STEPS

- Pilot with the stormwater team to gather real feedback
- Documenting work and procedures developed
- Handoff to multi-disciplinary the team can own it going forward

# REFLECTIONS

- Building a project from an empty canvas
- Figuring out the constraints and limitations of different frameworks
- Understanding the City's infrastructure and how our project will be part of it

# Thank you!

## Questions?

Special Thanks to  
Innovation Team  
Mo Spiesel, Stormwater GIS Analyst  
Gelcys Williams, Senior GIS Analyst